

LETTER TO THE EDITOR

Reply to: Comments on “Enhanced Diagnostic Accuracy of High-Grade Cervical Intraepithelial Neoplasia in Postmenopausal Women Through PAX1/JAM3 Methylation Analysis”

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Dear Editor,

We sincerely thank Fariha Shahid Tanveer and colleague for their thoughtful and constructive comments regarding our article, “Enhanced diagnostic accuracy of high-grade cervical intraepithelial neoplasia in postmenopausal women through PAX1/JAM3 methylation analysis.” They accurately pointed out that postmenopausal women are diagnostic blind spots in cervical cancer screening due to cervical atrophy and the presence of type 3 transformation zone (TZ3), which can compromise the performance of cytology and colposcopic evaluation [1]. We appreciate the opportunity to clarify key aspects of our study and to outline future directions.

Our study was designed as a colposcopy-based evaluation, and we fully agree that this has implications for generalization. As clearly stated in our Abstract and Methods, we enrolled women aged ≥ 50 years who underwent colposcopy due to positive screening results and/or abnormal clinical symptoms, rather than an unselected screening population. This referral-based design enriches the prevalence of high-grade disease and may therefore upwardly bias apparent sensitivity and specificity compared with population-based screening settings. Accordingly, in our Discussion we explicitly acknowledged potential selection bias related to referral to colposcopy and differences in screening behavior/access to care, and we cautioned that generalizability

should be interpreted carefully. Importantly, our primary objective was to evaluate the diagnostic-support/triage value of PAX1/JAM3 methylation among women already entering diagnostic evaluation pathways—an especially relevant context for postmenopausal women facing TZ3/atrophy-related limitations in cytology and colposcopy. We fully concur that a key next step is validation in population-based or prospective screening cohorts with lower disease prevalence and a different spectrum of pathology.

In addition, we agree that our cross-sectional design limits inference regarding risk prediction and lesion progression. Our study cannot directly evaluate future risk, lesion persistence, or progression, and we explicitly noted that “our cross-sectional design does not allow direct measurement of HPV persistence duration” and that the lack of longitudinal data limits causal inference. At the same time, we aimed to minimize misinterpretation by anchoring our conclusions to diagnostic/triage performance against pathology-based endpoints (colposcopy-directed biopsy/ECC and, when indicated, excisional procedures), rather than suggesting prognostic utility. The observed higher methylation levels in older women with advanced lesions were framed as hypothesis-generating and requiring prospective confirmation. We share the correspondents’ concern about distinguishing clinically meaningful disease from age-related epigenetic changes.

Abbreviations: CIN, cervical intraepithelial neoplasia; CIN2+, cervical intraepithelial neoplasia Grade 2 or worse; CISCER, PAX1/JAM3 methylation test; Ct, cycle threshold; ECC, endocervical curettage; HPV, human papillomavirus; hrHPV, high-risk human papillomavirus; JAM3, junctional adhesion molecule 3; LBC, liquid-based cytology; NMPA, China National Medical Products Administration; PAX1, paired box 1; TZ3, type 3 transformation zone.

In our dataset, methylation levels tracked lesion severity (Δ Ct progressively decreased with increasing CIN grade), and prolonged HPV persistence has been associated with increased methylation levels [2]. Additionally, we found no significant difference by HPV16/18 versus non-16/18 types, supporting a disease-severity linkage in this clinical context. Nevertheless, we agree that longitudinal follow-up in low-prevalence settings will be essential to address potential age-related epigenetic drift and to avoid overtreatment [3].

We also appreciate the methodological caution regarding the number of CIN2+ cases in the postmenopausal subgroup and the potential impact on precision. While our study represents one of the larger colposcopy-based postmenopausal cohorts evaluating combined PAX1/JAM3 methylation, we explicitly acknowledged that the absolute number of CIN2+ cases in the ≥ 50 group was limited ($n=44$ total CIN2+; CIN2=8, CIN3=15, SCC=18, AD=3), which reduces precision and generalizability and may widen confidence intervals. Regarding overfitting, our analysis evaluated a pre-specified commercial assay with pre-established Δ Ct cutoffs (PAX1 ≤ 6.6 ; JAM3 ≤ 10.0) supported by prior protocols [4, 5], rather than developing a new model or data-driven threshold within this cohort, which reduces overfitting risk. We fully agree that larger, multi-center cohorts with more CIN2/3 cases and richer stratification (HPV genotype, prior screening history, transformation zone status) are needed, and we also recommend future studies with systematic capture of prior screening history and larger multi-center sample sizes.

From a translational perspective, we strongly agree that the immediate utility of this assay lies in triaging borderline cases (e.g., hrHPV-positive/cytology-negative women, or those with inaccessible transformation zones) [6]. In our cohort, traditional strategies in postmenopausal women showed notable false positives/negatives, whereas CISCER (PAX1/JAM3 methylation) demonstrated high specificity ($\sim 93.6\%$) with high sensitivity ($\sim 93.2\%$) for CIN2+ in women ≥ 50 years, outperforming cytology and the guideline-based combined approach in specificity. We also provided triage analyses indicating that adding CISCER could reduce false positives after HPV stratification and reduce false negatives in cytology-based pathways. Clinically, we highlighted value in scenarios where conventional screening may miss disease: notably, two adenocarcinoma cases in our cohort were hrHPV-negative and cytology-negative, yet showed high methylation signals, suggesting methylation may complement HPV/cytology gaps in older women. We also described methylation-positive high-grade lesions among non-16/18 hrHPV-positive NILM cases [7, 8].

Finally, we agree that broader implementation requires external validation beyond a single regulatory context, and that cost-effectiveness and reproducibility across diverse settings remain important open questions. We reported that the assay used (CISCER DNA Methylation Detection Kit) is approved by the China National Medical Products Administration (NMPA) as a Class III medical device (No. 20233400253), and that cut-offs were established/validated through the registration process. However, we acknowledge differences in economic levels and screening strategies across countries and regions. The correspondents' emphasis on cross-system reproducibility and

health-economic evaluation is well taken. Our study focused on clinical performance in a colposcopy-referred setting and did not include formal cost-effectiveness analyses. Going forward, we agree that implementation research should include external validation in different healthcare systems and laboratory settings, inter-laboratory reproducibility studies with standardized quality control, and cost-effectiveness/budget-impact analyses aligned with local screening algorithms and resource constraints, as well as evaluation of workflows in older women with TZ3 where ECC/colposcopy sensitivity is challenged [9, 10].

In summary, we appreciate the correspondents' recognition of the clinical need in postmenopausal women and agree with their key methodological and translational recommendations. Our study demonstrates strong diagnostic-support performance of PAX1/JAM3 methylation in a colposcopy-based cohort of women ≥ 50 years and provides a rationale for its role as a triage/supplementary test in scenarios where HPV/cytology may be limited. We also transparently acknowledged limitations—including referral bias, lack of systematic prior screening history, cross-sectional design, and limited CIN2+ counts—and we concur that prospective population-based and longitudinal studies are the next necessary step.

Author Contributions

Huanzi Peng: conceptualization, methodology, validation, writing – original draft. **Jing Li:** conceptualization, methodology, formal analysis, writing – original draft. **Qun Zhou:** methodology, formal analysis. **Hui Zhou:** investigation, validation. **Jiaqi Peng:** validation, formal analysis, writing – original draft. **Jing Wang:** writing – original draft, formal analysis, methodology, conceptualization. **Pei Liu:** conceptualization, supervision, project administration, writing – review and editing. **Kun He:** conceptualization, writing – original draft, formal analysis. **Wene Liu:** investigation, methodology, writing – original draft. **Ping Tan:** investigation, validation. **Lei Li:** writing – review and editing, methodology. **Xiaobing Xie:** writing – review and editing, funding acquisition.

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Conflicts of Interest

The authors declare no conflicts of interest.

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